

FitBuddy – AI Fitness Plan Generator

Project Testing Phase – Page 1

Testing Phase Overview

Testing verifies that FitBuddy functions correctly according to its requirements and design.

- Check individual functions
- Check routes
- Check database
- Check AI integration
- Check complete workflow

Testing Objectives

The objective is to identify errors before final demonstration and deployment.

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Unit Testing

Individual functions are checked separately before full integration.

- Test workout generator
- Test nutrition function
- Test feedback update
- Test database functions
- Check input handling

Expected Result

Each function should return the expected type of result and handle normal inputs without unexpected errors.

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Input Testing

Form fields are tested using valid, incomplete and inappropriate values to verify validation and error handling.

- Age input
- Weight input
- Fitness goal
- Workout intensity
- Required fields

Validation Goal

The application should avoid processing incomplete or invalid information where validation is required.

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Route Testing

Each FastAPI route is tested separately.

- Test /
- Test /generate-workout
- Test /submit-feedback
- Test /view-all-users

Route Verification

The correct template, processing function and output should be associated with each route.

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AI Integration Testing

The Gemini integration is tested to verify that requests are sent correctly and generated responses are received and displayed.

- Check API configuration
- Check prompt input
- Check response handling
- Check failure handling

AI Output Testing

Generated plans are checked for the expected 7-day structure and relevance to the submitted fitness goal and intensity.

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Database Testing

Database operations are tested for creating, storing, retrieving and updating user and plan information.

- Save user
- Save plan
- Retrieve user
- Retrieve original plan
- Update plan

Data Consistency

Stored information should correspond to the user request and the generated or updated plan.

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Integration Testing

The complete flow is tested from user input to AI generation, database storage and result display.

- Input → route
- Route → AI
- AI → database
- Database → result
- Feedback → updated plan

End-to-End Scenario

A complete test uses a sample user, generates a plan, submits feedback and confirms that the revised plan is displayed.

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UI and Usability Testing

Pages are checked for readability, correct form controls, navigation and responsive presentation.

- Check labels
- Check buttons
- Check result formatting
- Check feedback form
- Check user view

Error Testing

The application is tested for missing API configuration, invalid requests, unavailable AI responses and database problems.

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Final Test Checklist

Before demonstration, all major functions should be executed successfully.

- Application starts
- Input page loads
- Workout generated
- Nutrition tip displayed
- Data stored
- Feedback updates plan
- All-users page works

Test Documentation

Test cases, expected results, actual results and corrections should be recorded for project documentation.

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Testing Summary

Testing confirms whether the implemented system behaves as expected and identifies issues requiring correction.

- Functional testing
- Integration testing
- UI testing
- Database testing
- Error testing

Testing Conclusion

After successful testing and correction of identified issues, FitBuddy is ready for final documentation and demonstration.